Forensic Baseline Report

Drone Forensic Decision and Orchestration Framework

Case Identifier:	CASE-df048-jun-ios-run2
Operator:	Floris
Analysis Started:	2026-06-16 11:47:25
Analysis Completed:	2026-06-16 11:50:51
Evidence Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\ios
Output Directory:	C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\dfdof_output\CASE-df048-jun-ios-run2
Evidence Sources:	3
Pipeline Version:	DFDOF v1.0 (proof of concept)

This report is an automated forensic baseline produced by DFDOF. It is not a court-ready expert report. All findings are bounded by the data available and must be interpreted by a qualified forensic examiner.The populated case output directory is the authorative source and provides the full overview of the pipeline's findings.

1. Evidence Intake and Chain of Custody

The following input evidence files were ingested. SHA-256 and SHA-1 hashes were computed at intake to establish integrity baselines. No modification to original evidence files is performed by DFDOF.

File	Classification	Acquisition	Size	Hash Timestamp
df048_external.E01	drone_sd	physical	7.8 GB	2026-06-16 11:47:50
df048_internal.001	drone_sd	physical	8.0 GB	2026-06-16 11:48:14
ios_logical.zip	controller_ios	logical	473.0 MB	2026-06-16 11:48:16

Evidence Source Path

C:\Users\Floris\Documents\dji_drones\train_mavic_air\df048\2018_june\ios

Cryptographic Hashes

df048_external.E01	drone_sd	physical
SHA-256:	`435b97c0cba3007d40b4ac10686049d1527fa0dc1235d8492a2df3e66d8bdac9`	
SHA-1:	`f46a295ef957df0dc3616ea5a95567f8df33e05b`	

df048_internal.001	drone_sd	physical
SHA-256:	`71b288d9b8e639a9b9f7c788dfe6bf71a148e8a4c93377b59cb36c35113a8ca6`	
SHA-1:	`8ae48fb181c3cf3c78e2e9680366f625d860a4a0`	

ios_logical.zip	controller_ios	logical
SHA-256:	`1d109ab0d67bebbc813e0fbfefa2f484af3805013db8483ff38f995ae509c820`	
SHA-1:	`5bed408f51e82fab79de2c7a30e4668ce58d85a4`	

2. Pipeline Execution Summary

Completed Phases

1. p1_provenance
2. p2_image_parsing
3. p3_artefact_extraction
4. p4_decision_and_orchestration
5. p5_normalisation_and_anomaly_checking
6. p6_multisource_correlation
7. p7_analysis_and_validation
8. p8_automated_reporting

Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_external.E01; treating as direct partition image
2. [p1 - drone sd]: mmls: no partition table found in df048_internal.001; treating as direct partition image
3. [p2 - controller android]: source evidence not found
4. [p3 - controller android]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - drone flight storage]: source evidence not found

Tool Execution Status

Tool	Invocations	Successes	Status
datcon	1	1	[OK]
exiftool	24	24	[OK]
fls	2	2	[OK]
icat	20	20	[OK]
mmls	2	0	[X] Failed
parser_ios	1	1	[OK]
txtlogtocsvtool	2	2	[OK]

Processing Coverage Score

Metric	Result
Evidence Sources Detected	2 / 4
Artefact Categories With Data	7 / 7
Flights With Primary Correlation	1 / 2
Tools Succeeded	6 / 7

3. Device and Account Identification

Field	Value	Confidence	Sources
Installed DJI App(s)	com.dji.go com.dji.pilot	[K] Controller-only	1
Controller Device	iPad mini 4	[K] Controller-only	1
Drone Serial Number	0K1DECE2BC0633	[K] Controller-only	3
Drone Model	Mavic Air	[K] Controller-only	4
Controller Device	iPad5,1__weibosdk__003143000__iphone__os 11.4	[K] Controller-only	2
DJI Application Version	3.1.38 4.2.20	[!] Inconsistent	4
Account E-Mail	droneforensics@vtoinc.com	[K] Controller-only	2
Product Type	38 7	[!] Inconsistent	2
Last Country Code	US	[K] Controller-only	2
Last App Launch	2018-06-19T17:40:02Z 2018-06-19T17:40:08Z	[!] Inconsistent	2
Last Flight Date	1970-01-01T00:00:00Z	[K] Controller-only	2
Last Known Latitude (Find Aircraft)	39.961218	[K] Controller-only	1
Last Known Longitude (Find Aircraft)	-106.21632703	[K] Controller-only	1
Last Known Altitude (Find Aircraft)	0.0	[K] Controller-only	1
Last Known Heading (Find Aircraft)	-149.7	[K] Controller-only	1
Last Known Horiz. Accuracy (Find Aircraft)	1.0	[K] Controller-only	1

4. Flight Analysis - Flight 01

Correlation and Flight Window

Status:	Matched (primary GPS+temporal)
Confidence:	medium
GPS Overlap:	175.7 s
Median GPS Distance:	6.28 m
Overlap:	100%
GPS Match Rate:	100%
Flight Record SHA-256:	`f81c1a6563583d3a4440a55510fa03d004c909adee313fd8c7f5cfbfac94882c`
Flight Record Name:	DJIFlightRecord_2018-06-19__12-25-59_.csv
Start:	2018-06-19T18:25:59.319000+00:00
End:	2018-06-19T18:28:55.017000+00:00
Duration:	175.7 s
Drone Log SHA-256:	`bcf4a8de1eb705a36dd27b56f92941c4603c47673aad5304c6accbd35456aa01`
Drone Log Name:	2018-06-19_12-20-49_FLY047.csv
Start:	2018-06-19T18:14:29.452000+00:00
End:	2018-06-19T18:35:22.087000+00:00
Duration:	1252.6 s

Flight Metrics

Peak Height:	16.6 m
Distance Travelled:	263.22 m
Video Recording:	No
Photo Capture:	No
Photos Taken:	0
Recording Duration:	0.0 s
Record Complete:	Yes

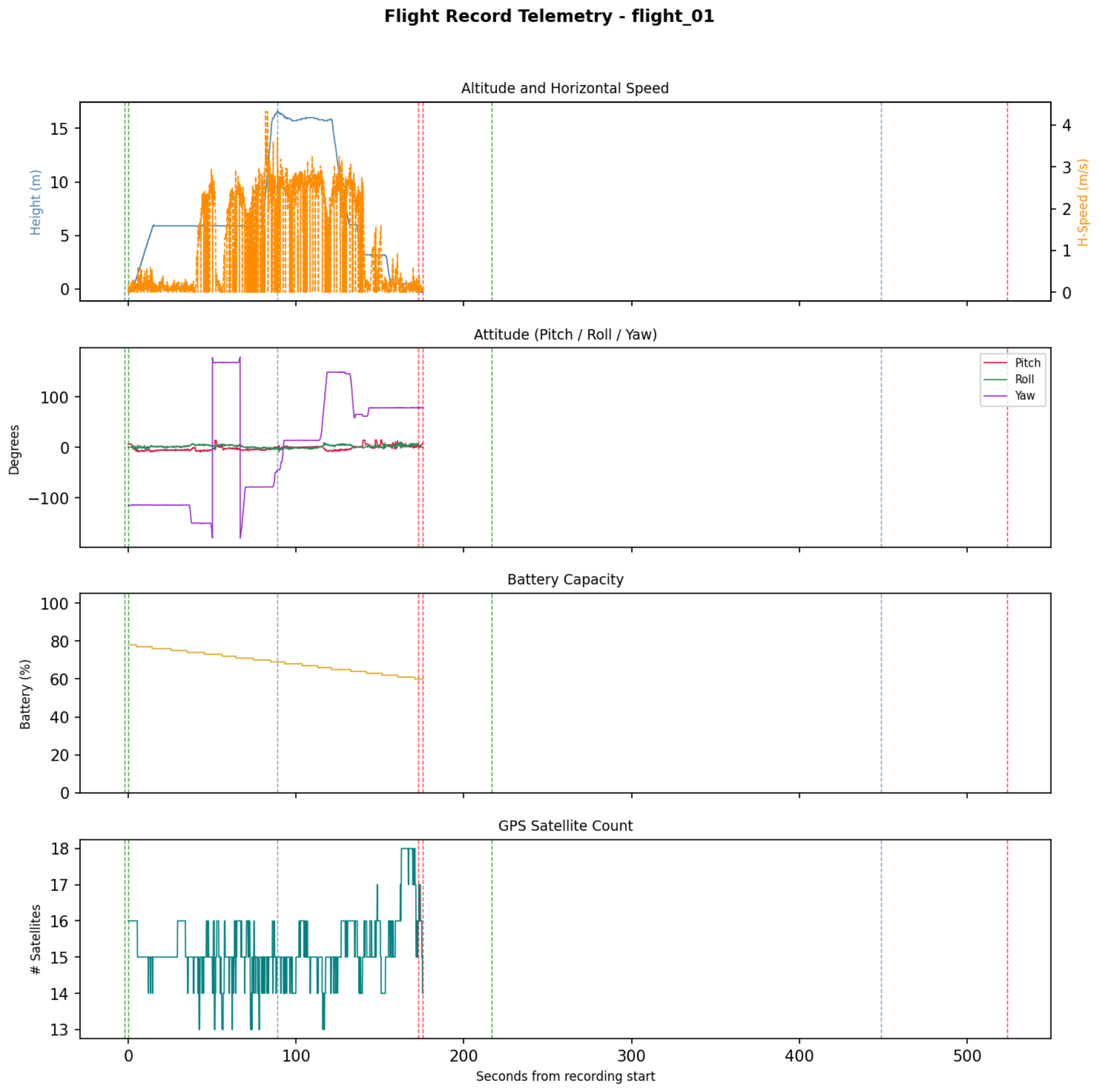
Fly Modes Observed (Chronological)

1. GPS_Atti
2. Atti
3. ASST_TAKEOFF
4. AutoTakeoff
5. EngineStart
6. AutoLanding

Key Events

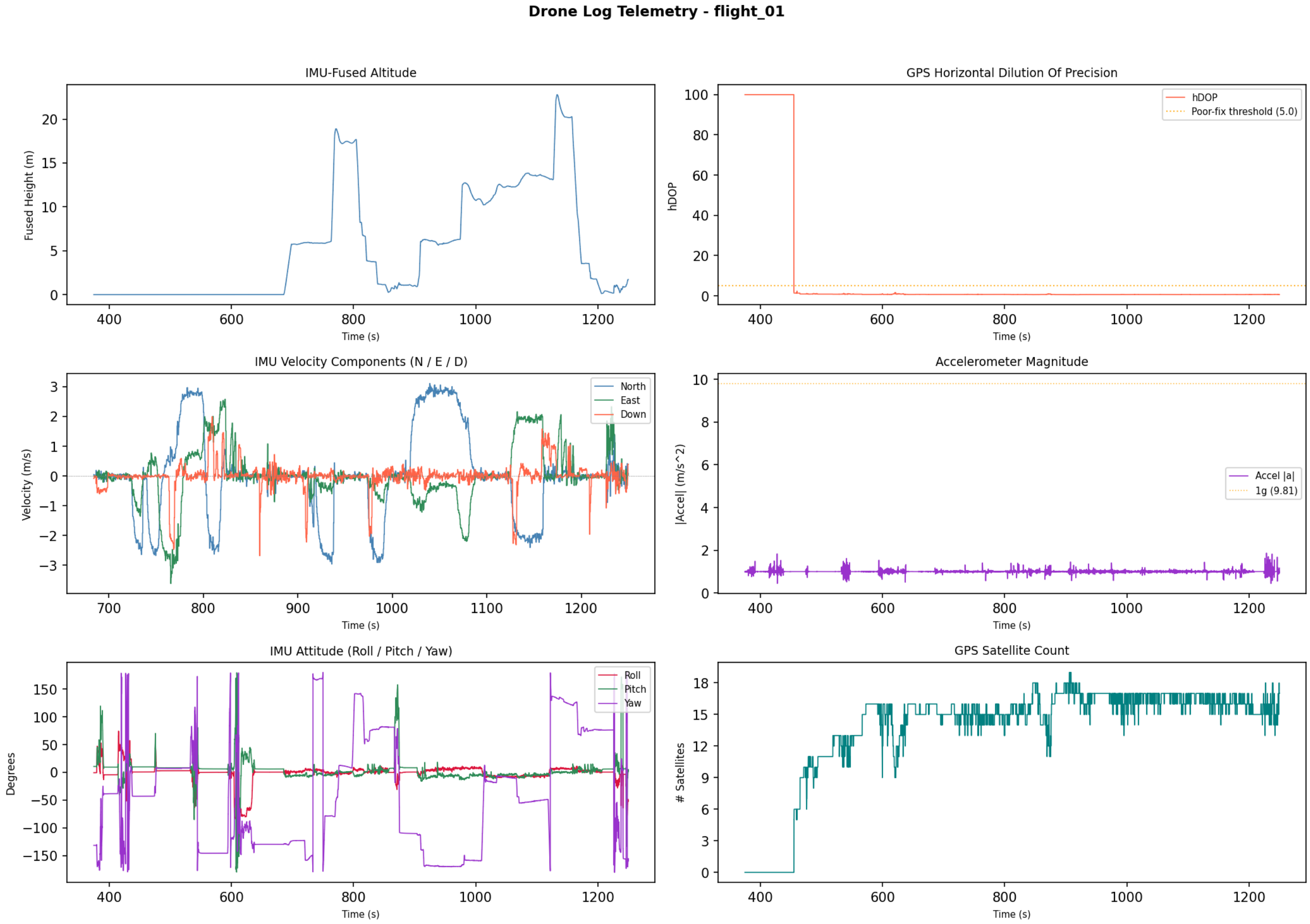
Timestamp (UTC)	Source	Event	Details
-	ctrl_ios:dl	Motor turned off	off
2018-06-19 18:22:10	ctrl_ios:dl	Log started	MavicAir
2018-06-19 18:25:57	ctrl_ios:dl	Motor turned on	on
2018-06-19 18:25:59	ctrl_ios:fr (v4.2.20)	Log started	SN: 0K1DECE2BC0633; App: 4.2.20; Mavic Air
2018-06-19 18:25:59	ctrl_ios:fr (v4.2.20)	Motor turned on	on
2018-06-19 18:25:59	ctrl_ios:fr (v4.2.20)	Record mode changed	No
2018-06-19 18:27:28	ctrl_ios:fr (v4.2.20)	Reached peak height	16.6 m
2018-06-19 18:28:52	ctrl_ios:dl	Motor turned off	off
2018-06-19 18:28:55	ctrl_ios:fr (v4.2.20)	Log ended	Dist home: 1.0 m; Height: 0.0 m; Rec: 0.0 s
2018-06-19 18:28:55	ctrl_ios:fr (v4.2.20)	Motor turned off	off
2018-06-19 18:29:36	ctrl_ios:dl	Motor turned on	on
2018-06-19 18:33:28	ctrl_ios:dl	Reached peak height	22.8 m
2018-06-19 18:34:43	ctrl_ios:dl	Motor turned off	off
2018-06-19 18:35:24	ctrl_ios:dl	Log ended	

Flight Record - Sensor Telemetry



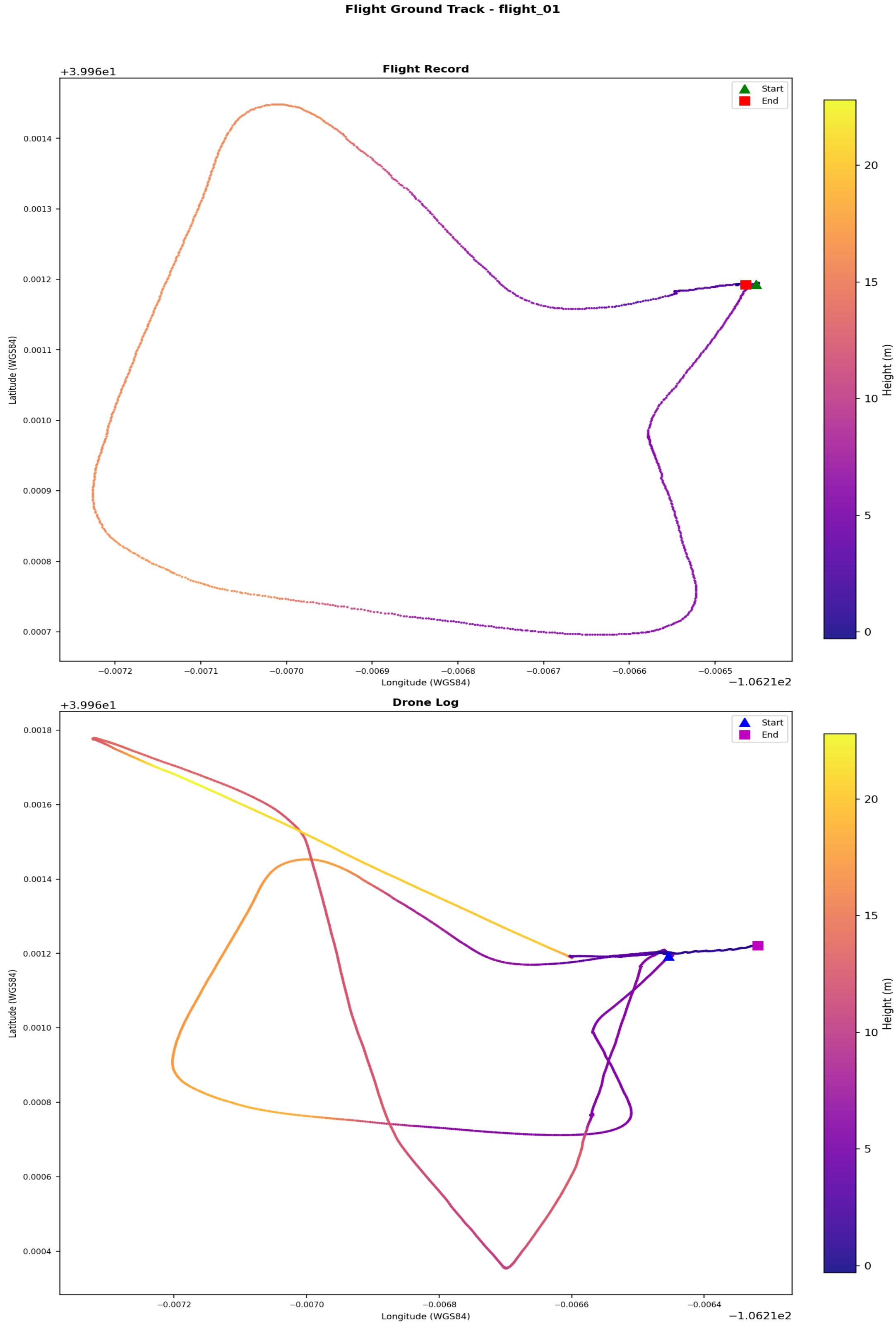
GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Drone Log - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver; horizontal accuracy is subject to satellite geometry (hDOP) and signal environment, with DJI manufacturer specification of +/-1.5 m horizontal in GPS mode. An hDOP below 1.5 is considered very precise, while values above 5.0 indicate degraded positional accuracy - those records are flagged and should be treated with reduced confidence in derived locations and ground track. Bayesian likelihood ratio analysis for a specific location hypothesis is possible using a Gaussian error model parameterised by hDOP, but requires case-specific inputs and is outside the scope of this baseline report.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (m). | KML: 2018-06-19_12-20-49_FLY047.kml

Coordinates are WGS84. No base map. Colour represents relative height (metres).

A KML file (2018-06-19_12-20-49_FLY047.kml) is available in the output directory and may be loaded into Google Earth Pro for 3D visualisation of the flight path.

Note: loading this file into an external application constitutes forensic evidence disclosure - ensure this is authorised before doing so.

4. Flight Analysis - Flight 02

Correlation and Flight Window

Status:	Unmatched (no drone log corroboration)
Flight Record SHA-256:	`933a9e3b9c25301aa60f86650614f7e4bc8c1bb472dce60ebd0ee65690f545f0`
Flight Record Name:	DJIFlightRecord_2018-06-19__12-29-38_.csv
Start:	2018-06-19T18:29:38.294000+00:00
End:	2018-06-19T18:34:45.015000+00:00
Duration:	306.7 s

Flight Metrics

Peak Height:	19.8 m
Distance Travelled:	399.66 m
Video Recording:	No
Photo Capture:	No
Photos Taken:	0
Recording Duration:	0.0 s
Record Complete:	Yes

Fly Modes Observed (Chronological)

1. EngineStart
2. AutoTakeoff
3. GPS_Atti
4. AutoLanding

Key Events

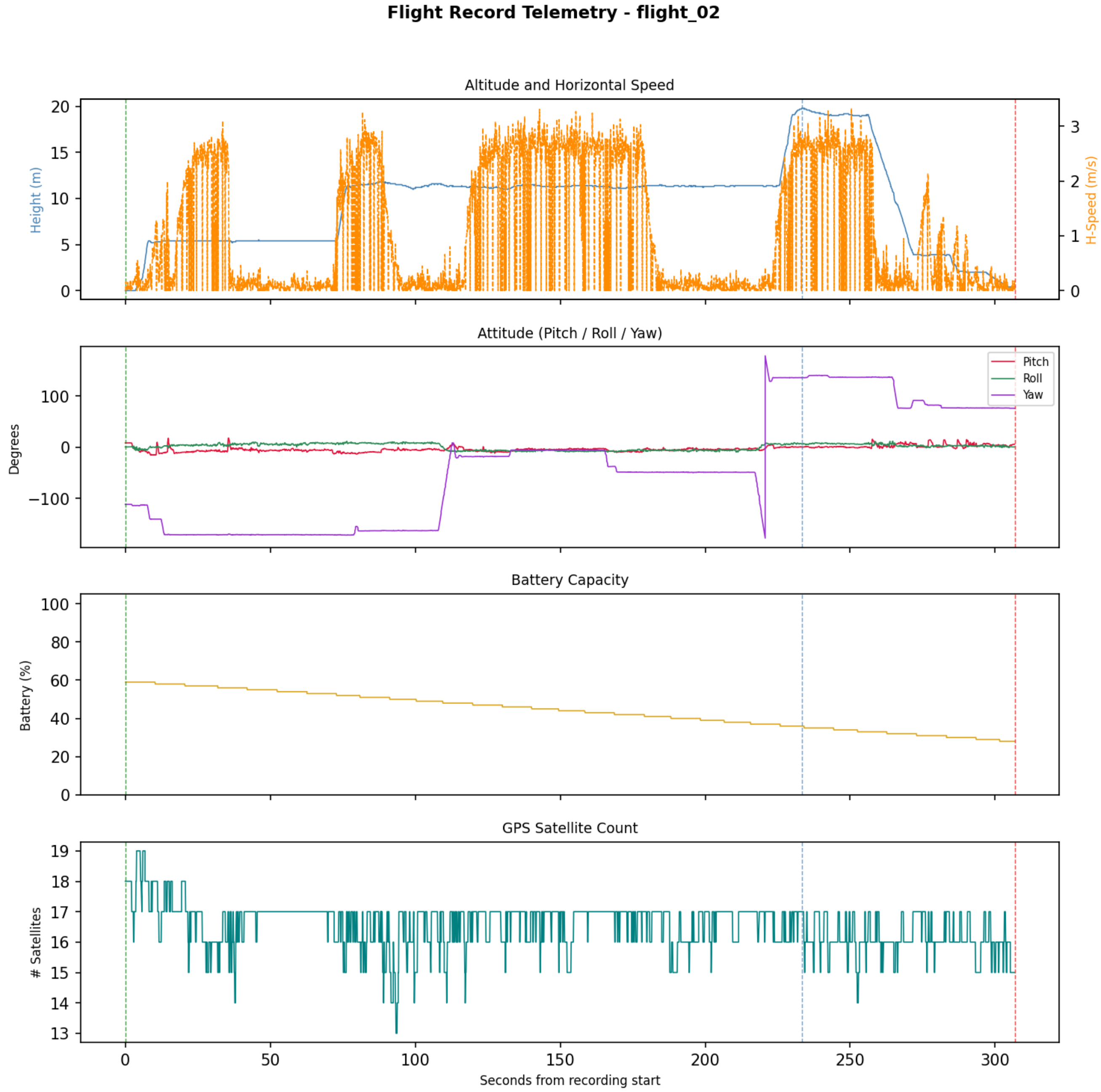
Timestamp (UTC)	Source	Event	Details
2018-06-19 18:29:38	ctrl_ios:fr (v4.2.20)	Log started	SN: 0K1DECE2BC0633; App: 4.2.20; Mavic Air
2018-06-19 18:29:38	ctrl_ios:fr (v4.2.20)	Motor turned on	on
2018-06-19 18:29:38	ctrl_ios:fr (v4.2.20)	Record mode changed	No
2018-06-19 18:33:31	ctrl_ios:fr (v4.2.20)	Reached peak height	19.8 m
2018-06-19 18:34:45	ctrl_ios:fr (v4.2.20)	Log ended	Dist home: 0.8 m; Height: 0.4 m; Rec: 0.0 s
2018-06-19 18:34:45	ctrl_ios:fr (v4.2.20)	Motor turned off	off

Correlated Artefacts

Possibly correlated (indirect indicators):

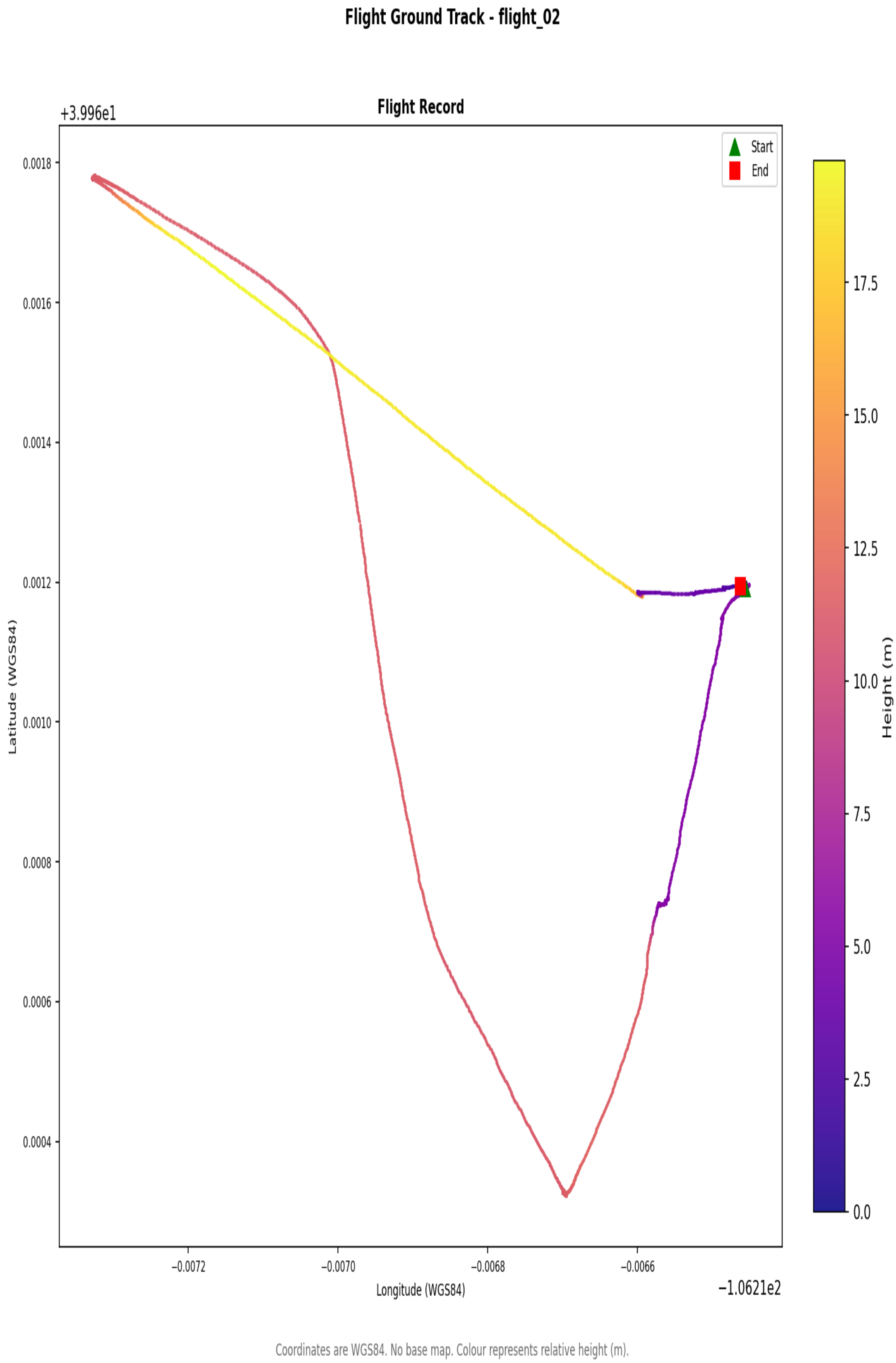
controller_ios: flight_logs	Log messages match (4 entries, format: json_array_log)
p5:2f8176e95fd6e0451726c8c23af817bd0133448a88c6b3b171f63eda826380c2	

Flight Record - Sensor Telemetry



GPS coordinates are derived from the drone's onboard receiver and represent sensor-recorded positions, not independently verified measurements. A satellite count of 10 or more is associated with good horizontal precision (approximately 1.5-3 m); counts below 4 indicate a poor fix with significantly reduced positional reliability, and those rows are flagged in this report. Low satellite counts may affect the accuracy of the flight path, ground track, and any location-based conclusions drawn from this record.

Flight Ground Track



Coordinates are WGS84. No base map. Colour represents relative height (metres).

5. Artefact Coverage

Source Coverage

Source	Detected
controller_android	No
controller_ios	Yes
drone_sd	Yes
drone_flight_storage	No

Artefact Counts per Source and Category

Source	Account Data	Databases	Drone Logs	Flight Logs	Flight Records	Images	Videos
controller_ios	2	14	1	5	2	2	2
drone_sd	0	0	0	0	0	10	10

Data Quality Notes per Category

Databases (14 files):

- 7 database(s) empty (controller_ios)

Drone Logs (1 files):

- 1 file(s) with row anomalies
- 1 file(s) with empty columns
- 1 file(s) with constant-value columns

Flight Logs (5 files):

- formats: 4 bracketed_log, 1 json_array_log

Flight Records (2 files):

- 1 file(s) with row anomalies
- 2 file(s) with empty columns
- 2 file(s) with constant-value columns

Images (12 files):

- 12 file(s) missing (normalisable) EXIF timestamp
- 12 file(s) missing EXIF GPS coordinates

Videos (12 files):

- 12 file(s) missing (normalisable) EXIF timestamp
- 12 file(s) missing EXIF GPS coordinates

6. Data Quality and Anomalies

Data Quality Observations

- Drone_logs: 1 file(s) with row anomalies detected in telemetry data.
- Drone_logs: 1 file(s) with empty columns - some telemetry channels contain no data.
- Drone_logs: 1 file(s) with constant-value columns - some telemetry channels may be unreliable.
- Flight_logs: log files present in the following format(s): 4 bracketed_log, 1 json_array_log.
- Flight_records: 1 file(s) with row anomalies detected in telemetry data.
- Flight_records: 2 file(s) with empty columns - some telemetry channels contain no data.
- Flight_records: 2 file(s) with constant-value columns - some telemetry channels may be unreliable.

Pipeline Anomaly Flags

1. [p1 - drone sd]: mmls: no partition table found in df048_external.E01; treating as direct partition image
2. [p1 - drone sd]: mmls: no partition table found in df048_internal.001; treating as direct partition image
3. [p2 - controller android]: source evidence not found
4. [p3 - controller android]: source evidence not found
5. [p3 - drone flight storage]: source evidence not found
6. [p4 - controller android]: source evidence not found
7. [p4 - drone flight storage]: source evidence not found

Artefact-Level Anomaly Detail

The following summarises which automated checks fired for each artefact processed in Phase 5. Row counts indicate affected data rows; column lists are truncated to five entries.

Databases

5255DF85724869CD36D4AFCD12AE092D.db

- database_empty: all tables empty

djiFMDB.db

- database_empty: all tables empty

djiFMDB_1.db

- database_empty: all tables empty

flysafe_app_dynamic_areas.db

- database_empty: all tables empty

flysafe_app_dynamic_areas_1.db

- database_empty: all tables empty

flysafe_dji_flight_dynamic_areas.db

- database_empty: all tables empty

flysafe_dji_flight_dynamic_areas_1.db

- database_empty: all tables empty

Drone Logs

norm_2018-06-19_12-20-49_FLY047.csv

- timestamp_gap: 1 row(s) affected
- missing_gps: 7 row(s) affected
- motor_airborne_off: 683 row(s) affected
- gps_poor_fix: 1659 row(s) affected
- contains_no_value: 6 column(s) - 4.3.0, ConvertDatV3, IMU_ATTI(0):absoluteHeight:C, IMU_ATTI(0):distanceHP:C, IMU_ATTI(1):absoluteHeight:C (+1 more)
- contains_constant_value: 43 column(s) - ATTI_MINI0:s_rsv00, ATTI_MINI0:s_rsv10, AirSpeed:vel_level:D, BatteryInfo:BatAuth:D, BatteryInfo:FullCap:D (+38 more)
- gps_accuracy_poor: 1659 row(s) affected
- attitude_out_of_bounds: 378 row(s) affected

Flight Logs

2018-06-19_12_26_01-0K1DECE2BC0633.dat

- format: json_array_log | entries: 4

upgradeLog_0.txt

- format: bracketed_log | entries: 6

upgradeLog_0_1.txt

- format: bracketed_log | entries: 6

upgradeLog_1.txt

- format: bracketed_log | entries: 4

upgradeLog_1_1.txt

- format: bracketed_log | entries: 2

Flight Records

norm_DJIFlightRecord_2018-06-19__12-25-59_.csv

- altitude_negative: 27 row(s) affected
- contains_no_value: 58 column(s) - APP_GPS.accuracy, APP_GPS.latitude, APP_GPS.longitude, APP_SER_WARN.warn, APP_WARN.warn (+53 more)
- contains_constant_value: 162 column(s) - APP_TIP.tip, CAMERA_INFO.cameraType, CAMERA_INFO.connectState, CAMERA_INFO.enabledPhoto, CAMERA_INFO.encryptStatus (+157 more)

norm_DJIFlightRecord_2018-06-19__12-29-38_.csv

- contains_no_value: 54 column(s) - APP_SER_WARN.warn, CENTER_BATTERY.connStatus, CENTER_BATTERY.errorType, CENTER_BATTERY.isBatteryOnCharge, CENTER_BATTERY.isNeedStudy (+49 more)
- contains_constant_value: 161 column(s) - APP_GPS.accuracy, APP_TIP.tip, APP_WARN.warn, CAMERA_INFO.cameraType, CAMERA_INFO.connectState (+156 more)

Images

2018_06_19_12_23_55.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2018_06_19_12_29_53.thumbnail

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp

- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.THM

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

Videos

2018_06_19_12_23_55.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

2018_06_19_12_29_53.mp4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0001.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0002.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0003.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0004.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

DJI_0005.MP4

- exif_contains_no_norm_time: is/are missing (normalisable) EXIF timestamp
- exif_contains_no_gps: is/are missing EXIF GPS coordinates

7. Further Investigation Items

The following items were identified by automated analysis as requiring manual investigator follow-up. They represent unresolved questions or limitations that the pipeline cannot resolve automatically.

1. DJI Application Version inconsistency detected - conflicting values observed: 3.1.38, 4.2.20. Manual verification required.
2. Product Type inconsistency detected - conflicting values observed: 38, 7. Manual verification required.
3. Note: no firmware version found - Device and account information.
4. Last App Launch inconsistency detected - conflicting values observed: 2018-06-19T17:40:02Z, 2018-06-19T17:40:08Z. Manual verification required.
5. 1 flight record(s) could not be correlated to a drone log (flight_02). Independent corroboration is absent for these flight(s).
6. 7 database file(s) contain records and should be examined for relevant artefacts.

Appendix A - Extracted Artefact Hash Manifest

Each artefact extracted in Phase 3 is listed below with its full SHA-256 and SHA-1 cryptographic hashes. These values can be used to independently verify the integrity of every extracted file.

com.dji.go.plist	Account Data	controller_ios	extract_logical
SHA-256:	7016ee7a1c90ff783060672ec5cfc5dad1d5fd75a67550fc4fc3fffe372ced92		
SHA-1:	c1c0878212248d16588ad0334b7873eb8e57edbd		

com.dji.pilot.plist	Account Data	controller_ios	extract_logical
SHA-256:	1e31cc73c9c096dd92895e2aed99a98c8fbe94cb7b7492ec9cd8732c98ab4025		
SHA-1:	d74d26eb07bdd20b77f76eb8bebb9f950d32040f		

5255DF85724869CD36D4AFCD12AE092D.db	Databases	controller_ios	extract_logical
SHA-256:	aae50532bee18355508de941e178ea847cd725018bb03a93589944f7cad4898e		
SHA-1:	7ecb7ed8a4b0ecf3028259a0eb0dcee998fcbc0c		

538103E3AB6E8F987EC10913E64A3E24.db	Databases	controller_ios	extract_logical
SHA-256:	6df74165eb44a8570bdc89f0038e0831de6ba509aa2aa2bd15aa1a6a2786cea0		
SHA-1:	f2d9e8167faa28bbb2685b330a7fabe2732943aa		

djiFMDB.db	Databases	controller_ios	extract_logical
SHA-256:	5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfecc749fd52a		
SHA-1:	b1d200f2a2416e3c776ebc4e1779312198adf52b		

djiFMDB_1.db	Databases	controller_ios	extract_logical
SHA-256:	5fdae73b94d5e7a078a8bfd1afda762eae9a5bff869139e9242dfecc749fd52a		
SHA-1:	b1d200f2a2416e3c776ebc4e1779312198adf52b		

dji_fs_upgrade_local_cache.db	Databases	controller_ios	extract_logical
SHA-256:	00dadc7c58ac4622096b9689e8d8d55b4fc63d197fd3672cf911032956459460		
SHA-1:	76ca4596009995c07cdec3ca79a06f7d8cd2d28a		

flysafe_app_dynamic_areas.db	Databases	controller_ios	extract_logical
SHA-256:	483da04ac8232310746927e831150673403cb79e12dbedf322869a1317fa7fcf		
SHA-1:	4e8e284810efa83aa0861217939fff02c9d7ad12		

flysafe_app_dynamic_areas_1.db	Databases	controller_ios	extract_logical
SHA-256:	ce02ae43e9061d526429d75f4aab106e9dc0133a90ac005df05e984ccf52e6fe		
SHA-1:	5c25a94e52815ddf092eccb33456a159cc66f577		

flysafe_areas_djigo.db	Databases	controller_ios	extract_logical
SHA-256:	b5a6d15e6e04e9cbcc0902a392540a2b25d2a9722c6219d5cf34dfad90f838c8		
SHA-1:	a2836e85087f06b331fbfea45025b05cf35f5942		

flysafe_areas_djigo_en.db	Databases	controller_ios	extract_logical
SHA-256:	1a1c024072fba37bffbf1198f4227fcf2f4ffad347ad9a93b164a7077cd0b6f8		
SHA-1:	761bad6393f8f8a28ecea57069626b9a9eaa6de4		

flysafe_dji_flight_dynamic_areas.db	Databases	controller_ios	extract_logical
SHA-256:	6d434e165f689c505c6d051cc3ba0aa7dd26737023ff1aee562fd88f106a8953		
SHA-1:	80d3466fd6360f9edab821a05cfbf97ac0ce6804		

flysafe_dji_flight_dynamic_areas_1.db	Databases	controller_ios	extract_logical
SHA-256:	56ef148210fceffbca82dc86b201fbdc340f32c9a247b9341ba85c54a21918c6		
SHA-1:	91649bae84edd1c230392932c2a5ad0e917b0f7c		

flysafe_license_unlock.db	Databases	controller_ios	extract_logical
SHA-256:	12c8a85f84d046cd254fccf764723a09d903c4d5d27cedb5fc9ba6f0077f7c44		
SHA-1:	ad4b118fc3dc923a8345d117abfb75c667a2a658		

flysafe_static_circle.db	Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4		
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4		

flysafe_static_circle_1.db	Databases	controller_ios	extract_logical
SHA-256:	4787fc859c865dab6b3d0db8eb68a55c764ca5d13105b14429cc2d455708cfb4		
SHA-1:	ccf2d5995e798abe1f1255c252c28ec8ae699dd4		

2018-06-19_12-20-49_FLY047.DAT	Drone Logs	controller_ios	extract_logical
SHA-256:	d654bc25090e6f85e056a3b4c6c8b2c063f56bc62d294f0abd24565e46a3f0fd		
SHA-1:	911c0b5b5830defa2a6bc96615a261670a48d87e		

2018-06-19_12_26_01-0K1DECE2BC0633.dat	Flight Logs	controller_ios	extract_logical
SHA-256:	2f8176e95fd6e0451726c8c23af817bd0133448a88c6b3b171f63eda826380c2		
SHA-1:	210076ad81bec9b1fdc92248b1ad12d4c3139469		

upgradeLog_0.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	448681afddf995a774589cdcbee50b298968a8814988b75953e420c78682fd57		
SHA-1:	1fd8cb99a20387423ee4a6a81c6ce240f000388f		

upgradeLog_0_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	9e8e8593e2320a11f72fd116131acd8a9b084675bfc44e52858809dad296ab94		
SHA-1:	cefc6b884dae38ef77a3bf0ea389493596c3073d		

upgradeLog_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	2ed69c816a961c09cf89529eca1ff77d15790292bb4ac6961fbc62b68b24aa49		
SHA-1:	f852f87d4e15162081c5a256a217fb7c5735c7ea		

upgradeLog_1_1.txt	Flight Logs	controller_ios	extract_logical
SHA-256:	d59332f1f2d8370a8eb739165b6b7af115e6ecbe7fe987431a2bffbf5f016e00		
SHA-1:	e87a6c2613770830b1f56bd2695c5ce31db19b80		

DJIFlightRecord_2018-06-19__12-25-59_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	f81a98ad5fc3c4c827810faae5eb6a26edbcfec0b196aafd054dacfc8f05bd70		
SHA-1:	edf8cebc08f9a9ef79de9a8aacf340ed4b18044b		

DJIFlightRecord_2018-06-19__12-29-38_.txt	Flight Records	controller_ios	extract_logical
SHA-256:	10cf71479b9c46a5c955f365c831e8b75efe60b93ca5218a7f4bbcb96ccecfee		
SHA-1:	02a7d82eaf9446a4e35ef83eb084f8dbab29a138		

2018_06_19_12_23_55.thumbnail	Images	controller_ios	extract_logical
SHA-256:	e9f76920c6d64ffe9d61cd8f39436c883a3a1e2e5864753e68e3ce7ad544b90d		
SHA-1:	c2efa6cb95262d41d87ede78731410b5831db489		

2018_06_19_12_29_53.thumbnail	Images	controller_ios	extract_logical
SHA-256:	af047a2532151dd31d241f564a8d075833fbb5cb6525cb3c64d1eb592f3c357b		
SHA-1:	5e5cc6cd69292e1938f7f832b900140f5647a1ae		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0		
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f		
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79		
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4		
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f		
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324		

DJI_0001.THM	Images	drone_sd	extract_physical
SHA-256:	3a3918e0c00dddb9ad706491020da1b7db90479f49e29814274087259da2bcf0		
SHA-1:	b6b5af5e4d4b0d5ecae0eef6e7901ec873ce8048		

DJI_0002.THM	Images	drone_sd	extract_physical
SHA-256:	c452f70bc9e9bad9f05047d07cf685498ab5117ec940a5bc6622bd77baa79b1f		
SHA-1:	9f15bf98c1c26619f2499ece191f0dc426aae6e3		

DJI_0003.THM	Images	drone_sd	extract_physical
SHA-256:	9727e9310aa901a379b5fdfaf29a50e22fb879ff86d7ba6d75e82c6a39497d79		
SHA-1:	c958de01cff11f28d30a028a327b0d9867c968fc		

DJI_0004.THM	Images	drone_sd	extract_physical
SHA-256:	f99c4b6f1f614bd2ea9829aa50e08d966ed188dde73ce9dfcf388ee70adb7dc4		
SHA-1:	434ae634442c15bfbe0b51dbc2b31193f14968d5		

DJI_0005.THM	Images	drone_sd	extract_physical
SHA-256:	b3f920d7ad5412a19b8f122aaa0ff4cedfcf9b25c2baa89bbf4d26e0d796fe5f		
SHA-1:	6f4b46f652887ad89b44020dc176a4ac9f0bb324		

2018_06_19_12_23_55.mp4	Videos	controller_ios	extract_logical
SHA-256:	31a8536319cbd593e8c0a33cec064b89099e2b22eec4cd95de7bdbe3904812eb		
SHA-1:	ebd5e99d83cce56f7274bdf792651052a749ebf7		

2018_06_19_12_29_53.mp4	Videos	controller_ios	extract_logical
SHA-256:	fa24dd461617c1472231e48ee9acbb7237c9348a563d564204e7999a021d3f40		
SHA-1:	7aa9722b3042ac9711b2e772033fee215ad85d90		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815		
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea		

DJI_0002.MP4	Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b		
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f		

DJI_0003.MP4	Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a		
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beaacb3		
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192		
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821		

DJI_0001.MP4	Videos	drone_sd	extract_physical
SHA-256:	61a290fa102ce3e2913242b83aae8073af3751cd6b8f1ed0e14eab7625f77815		
SHA-1:	9873e6568f8631e23ae819daa214f6e88312efea		

DJI_0002.MP4	Videos	drone_sd	extract_physical
SHA-256:	9db5019a6c22e8ce2c2d726df4f8c1c0b52fc3d5806b38058bf1896f80e6ba7b		
SHA-1:	5131359773d614150ff1d62d8ee4c4d984eb640f		

DJI_0003.MP4	Videos	drone_sd	extract_physical
SHA-256:	aad026be3d81c7dbf251c37d7be4776d239dd44cd7a9988190e6a0f04b87796a		
SHA-1:	e74bf86c588742fb6cf85fda3b641c41e7834b65		

DJI_0004.MP4	Videos	drone_sd	extract_physical
SHA-256:	6aa7dc8ae4f82d9238d99673a3b4c30ab87f3dfe46627de30b2d6f200beaacb3		
SHA-1:	8a6bfb4eca3908e47a7bab171aa907e333ed0ab7		

DJI_0005.MP4	Videos	drone_sd	extract_physical
SHA-256:	890e7c07f2d747cdbdffe8939636fc1c4f1230ce592db8d91fff8742551f8192		
SHA-1:	a262dd9af22576b92269280cc9e1fe567e375821		

Appendix B - Evidence Derivation Summary

This table summarises how many artefacts were derived at each pipeline phase, per evidence source and category. It demonstrates the processing chain from extracted artefacts (P3) through decoded outputs (P4) to normalised analysis-ready files (P5).

Controller Ios

Category	P3 Extracted	P4 Decoded	P5 Normalised
Account Data	2	2	0
Databases	14	0	0
Drone Logs	1	2	1
Flight Logs	5	0	0
Flight Records	2	2	2
Images	2	2	2
Videos	2	2	0

Drone Sd

Category	P3 Extracted	P4 Decoded	P5 Normalised
Images	10	10	10
Videos	10	10	0

Appendix C - Tool Invocation Log

All external tool invocations executed during this analysis are recorded below, in chronological order. Each entry includes the timestamp, tool name, version, return code, full command, and output files produced.

2026-06-16 11:47:25 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_external.E01

2026-06-16 11:47:25 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_external.E01

2026-06-16 11:47:25 mmls vThe Sleuth Kit ver 4.15.0 rc=1

cmd: mmls.exe df048_internal.001

2026-06-16 11:47:26 fls vThe Sleuth Kit ver 4.15.0 rc=0

cmd: fls.exe -r -p df048_internal.001

2026-06-16 11:48:24 parser_ios v- rc=n/a

cmd: ios_logical.zip C:\Users\Floris\Documents\dfdof_output\CASE-df048-jun-ios-run2\p2_image_parsing\controller_ios_parsed

out: controller_ios_parsed

2026-06-16 11:48:29 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357829

out: DJI_0001.MP4

2026-06-16 11:48:35 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357830

out: DJI_0002.MP4

2026-06-16 11:48:49 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357831

out: DJI_0003.MP4

2026-06-16 11:49:05 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357832

out: DJI_0004.MP4

2026-06-16 11:49:15 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1357833

out: DJI_0005.MP4

2026-06-16 11:49:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358853

out: DJI_0001.THM

2026-06-16 11:49:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358854

out: DJI_0002.THM

2026-06-16 11:49:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358855

out: DJI_0003.THM

2026-06-16 11:49:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358856

out: DJI_0004.THM

2026-06-16 11:49:20 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_external.E01 1358857

out: DJI_0005.THM

2026-06-16 11:49:22 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357829 *out: DJI_0001.MP4*

2026-06-16 11:49:26 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357830

out: DJI_0002.MP4

2026-06-16 11:49:39 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357831

out: DJI_0003.MP4

2026-06-16 11:49:54 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357832

out: DJI_0004.MP4

2026-06-16 11:50:03 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1357833

out: DJI_0005.MP4

2026-06-16 11:50:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358853

out: DJI_0001.THM

2026-06-16 11:50:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358854

out: DJI_0002.THM

2026-06-16 11:50:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358855

out: DJI_0003.THM

2026-06-16 11:50:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358856

out: DJI_0004.THM

2026-06-16 11:50:08 icat vThe Sleuth Kit ver 4.15.0 rc=0

cmd: icat.exe -o 0 df048_internal.001 1358857

out: DJI_0005.THM

2026-06-16 11:50:32 datcon v4.3.0 rc=n/a

cmd: DatCon.4.3.0.exe 2018-06-19_12-20-49_FLY047.DAT

out: 2018-06-19_12-20-49_FLY047.csv, 2018-06-19_12-20-49_FLY047.kml

2026-06-16 11:50:33 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19__12-25-59_.txt DJIFlightRecord_2018-06-19__12-25-59_.csv

out: DJIFlightRecord_2018-06-19__12-25-59_.csv

2026-06-16 11:50:34 txtlogtocsvtool v2018-06-11 rc=0

cmd: TXTlogToCSVtool.exe DJIFlightRecord_2018-06-19__12-29-38_.txt DJIFlightRecord_2018-06-19__12-29-38_.csv

out: DJIFlightRecord_2018-06-19__12-29-38_.csv

2026-06-16 11:50:34 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_12_23_55.thumbnail

out: 2018_06_19_12_23_55_exif.json

2026-06-16 11:50:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_12_29_53.thumbnail

out: 2018_06_19_12_29_53_exif.json

2026-06-16 11:50:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_12_23_55.mp4

out: 2018_06_19_12_23_55_exif.json

2026-06-16 11:50:35 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n 2018_06_19_12_29_53.mp4 *out: 2018_06_19_12_29_53_exif.json*

2026-06-16 11:50:36 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-16 11:50:36 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM
out: DJI_0002_exif.json

2026-06-16 11:50:36 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-16 11:50:37 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-16 11:50:37 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-16 11:50:37 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4
out: DJI_0001_exif.json

2026-06-16 11:50:38 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4
out: DJI_0002_exif.json

2026-06-16 11:50:38 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4
out: DJI_0003_exif.json

2026-06-16 11:50:38 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4
out: DJI_0004_exif.json

2026-06-16 11:50:39 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4
out: DJI_0005_exif.json

2026-06-16 11:50:39 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.THM
out: DJI_0001_exif.json

2026-06-16 11:50:39 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.THM
out: DJI_0002_exif.json

2026-06-16 11:50:40 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.THM
out: DJI_0003_exif.json

2026-06-16 11:50:40 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.THM
out: DJI_0004_exif.json

2026-06-16 11:50:40 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.THM
out: DJI_0005_exif.json

2026-06-16 11:50:41 exiftool v13.57 rc=0
cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0001.MP4

out: DJI_0001_exif.json

2026-06-16 11:50:41 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0002.MP4

out: DJI_0002_exif.json

2026-06-16 11:50:41 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0003.MP4

out: DJI_0003_exif.json

2026-06-16 11:50:42 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0004.MP4

out: DJI_0004_exif.json

2026-06-16 11:50:42 exiftool v13.57 rc=0

cmd: exiftool.exe -a -u -g1 -s -ee -json -n DJI_0005.MP4

out: DJI_0005_exif.json

Appendix D - DFDOF Proof of Concept Overview

The Drone Forensic Decision and Orchestration Framework (DFDOF) is an automated 8-phase forensic pipeline for DJI drone evidence. It ingests images of the controller, drone flight storage and SD cards, then processes them into a structured analysis and report.

P1 Provenance and Integrity

Classifies input files (ZIP/E01/.001) by content signatures. Computes SHA-256 and SHA-1 for all input evidence. Presents a source classification summary for operator confirmation before processing begins.

Input: evidence directory. Output: classified evidence objects with hashes.

P2 Image Parsing

Makes all sources search-ready. Decrypts iOS backups into a domain directory tree. Extracts Android device metadata (DeviceInfo.xml, ApplicationInfo.xml, packages.list). Produces backup_info.json with device identifiers and installed DJI applications.

Input: raw evidence files. Output: parsed directory structures, backup_info.json.

P3 Artefact Extraction

Root-driven category extraction. Discovers DJI app roots by bundle-ID segment matching. Recurses into configured per-platform category paths and filters by extension. Rejects empty files. Each artefact is wrapped in an Evidence object with a parent hash linking it to its source.

Input: parsed directories. Output: categorised artefacts (flight records, drone logs, flight logs, databases, images, videos, account data).

P4 Decision and Orchestration

Dispatches each artefact category to the appropriate specialist tool: DatCon for .DAT drone logs (produces CSV + KML), TXTlogToCSV for flight record .txt files, ExifTool for images and videos, Python parsers for account data (.plist/.xml). Computes hashes for all tool outputs.

Input: extracted artefacts. Output: decoded CSVs, KML files, EXIF JSON, account data observations.

P5 Normalisation and Anomaly Checking

Normalises all CSVs to UTC timestamps and a unified schema. Applies 9 shared telemetry anomaly checks (timestamp regression/gaps, zero coordinates, GPS quality, altitude spikes, motor-airborne-off) plus format-specific checks for flight records (battery voltage, temperature) and drone logs (quaternion, hDOP). Detects empty databases, unreadable flight logs, and missing (normalisable) EXIF timestamps.

Input: decoded artefacts. Output: normalised CSVs, anomaly observations.

P6 Multi-Source Correlation

Correlates flight records with drone logs using two criteria: (1) temporal overlap >= 60 s AND >= 90% of the shorter flight's duration; (2) median haversine distance of nearest-neighbour GPS pairs <= 25 m with >= 5 matched pairs. Deduplicates FR candidates when multiple apps recorded the same flight. Builds a unified per-flight event timeline from all matched sources. Also correlates media files by EXIF timestamp, GPS bounding box, and video duration. In the correlation summary: Overlap is the temporal overlap as a percentage of the shorter flight's duration; GPS Match Rate is the fraction of flight-record GPS rows that found a spatially matched drone-log row within a +-2 s window.

Input: normalised CSVs. Output: timeline_flightXX.json per flight.

P7 Analysis and Validation

Aggregates results from all prior phases into a structured analysis. Resolves device identity (drone serial, model, app version, account email) across sources with confidence levels (controller-only / corroborated / inconsistent). Computes per-flight metrics (peak height, media capture, record completeness). Produces coverage scores and a grouped conclusions structure with a dedicated further_investigation section.

Input: all phase outputs. Output: analysis dict with conclusions in state.json.

P8 Automated Reporting

Generates this PDF forensic baseline report from all prior phase outputs. Produces sensor telemetry plots (altitude, speed, attitude, battery, GPS) from normalised CSVs, a dual-source flight track plot, and the structured appendices. All figures are also saved as standalone PNG files.

Input: state.json (all phase outputs). Output: PDF report + PNG plot files.

Disclaimer

DFDOF was developed by Floris Krijger, MSc Security and Network Engineering, University of Amsterdam, as a Master's thesis project (1 April 2026 - 30 June 2026). This proof of concept has been developed and tested on a targeted subset of the NIST VTO forensic drone dataset and is not a finished, production-ready system. Before use in any operational or legal context, further testing, validation across additional drone models and evidence types, and robustness improvements are required. The authors accept no liability for conclusions drawn from automated pipeline output without independent expert verification.